Mini-Actvity-3-Solutions.R

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```
##### Mini-Activity 3 Solutions #####
```

Before beginning addressing the questions, we start by setting up our environment.

```
# Load libraries
library(psych)
library(here)
```

```
## here() starts at /Users/gcrone14/Desktop/R Projects/PSYC3031-F26
```

```
library(ez)
```

```
## Registered S3 method overwritten by 'lme4':
##   method           from
##   na.action.merMod car
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x ggplot2::%+%()   masks psych::%+%()
## x ggplot2::alpha() masks psych::alpha()
## x dplyr::filter()  masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Import data
games <- read_csv(here::here("Data/videogametype.csv"))
```

```
## Rows: 20 Columns: 4
## -- Column specification --------------------------------------------------------
## Delimiter: ","
```

```
## chr (2): hyperactivity, game_type
## dbl (2): ID, nviolent_images
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
glimpse(games) # Inspect data
```

```
## Rows: 20
## Columns: 4
## $ ID              <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,~
## $ hyperactivity   <chr> "hyperactive", "hyperactive", "hyperactive", "hyperact~
## $ game_type       <chr> "violent game", "violent game", "violent game", "viole~
## $ nviolent_images <dbl> 10, 13, 12, 10, 11, 4, 6, 3, 4, 3, 8, 11, 7, 9, 10, 5,~
```

```
table(games$hyperactivity) |> names()
```

```
## [1] "hyperactive"     "not hyperactive"
```

```
table(games$game_type) |> names()
```

```
## [1] "prosocial"    "violent game"
```

```
# Clean the data
games_final <- games |>
  mutate(ID = factor(ID),
         hyperactivity = factor(hyperactivity),
         game_type = factor(game_type))
glimpse(games_final)
```

```
## Rows: 20
## Columns: 4
## $ ID              <fct> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16,~
## $ hyperactivity   <fct> hyperactive, hyperactive, hyperactive, hyperactive, hy~
## $ game_type       <fct> violent game, violent game, violent game, violent game~
## $ nviolent_images <dbl> 10, 13, 12, 10, 11, 4, 6, 3, 4, 3, 8, 11, 7, 9, 10, 5,~
```

1)

Since we're given two categorical IVs with a continuous DV, we'd run a two-way (2x2) ANOVA.

2)

The research question might be: "Do hyperactive children tend to have more violent images appear in their dreams, stories, and pictures than non-hyperactive children, and does this effect change based on whether children play violent or prosocial video games?"

3)

We can compute descriptive stats to see what the answer might be.

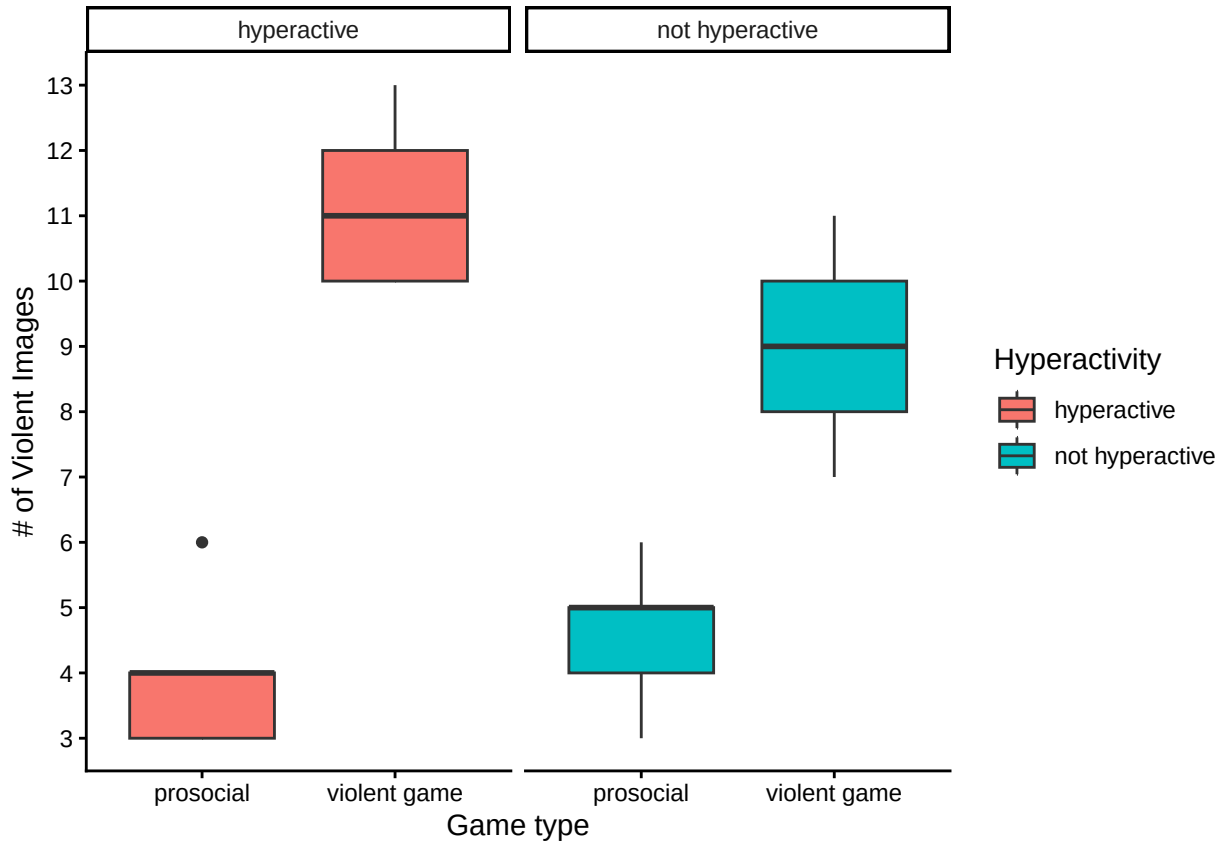
```r
games_final |>
  group_by(hyperactivity, game_type) |>
  summarise(mean_nviolent_images = mean(nviolent_images),
         sd_nviolent_images = sd(nviolent_images),
         median_nviolent_images = median(nviolent_images))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by hyperactivity and game_type.
## i Output is grouped by hyperactivity.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(hyperactivity, game_type))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 4 x 5
## # Groups:   hyperactivity [2]
##   hyperactivity   game_type    mean_nviolent_images sd_nviolent_images
##   <fct>           <fct>                       <dbl>              <dbl>
## 1 hyperactive     prosocial                     4                 1.22
## 2 hyperactive     violent game                 11.2               1.30
## 3 not hyperactive prosocial                     4.6               1.14
## 4 not hyperactive violent game                  9                 1.58
## # i 1 more variable: median_nviolent_images <dbl>
```

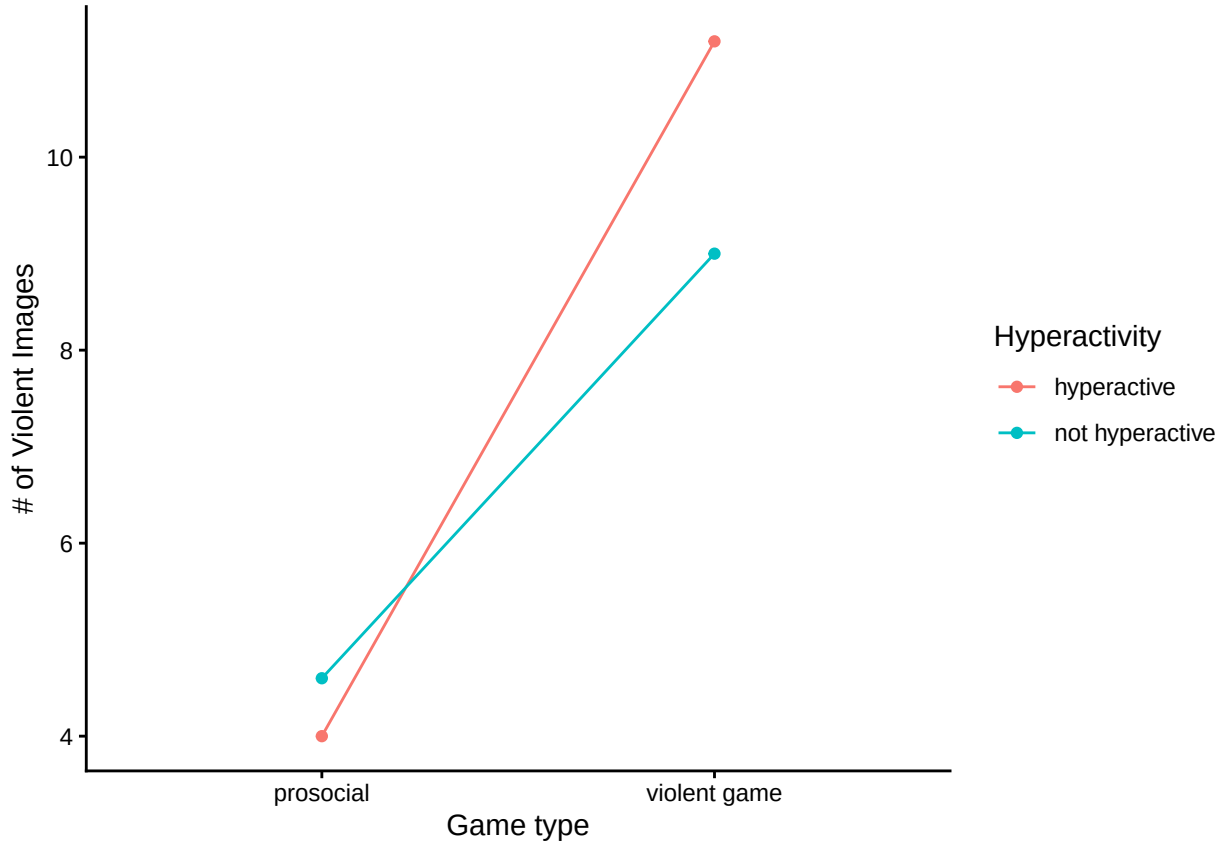
Let's plot this relationship.

```r
games_final |>
  ggplot(aes(x = game_type, y = nviolent_images,
             fill = hyperactivity)) +
  geom_boxplot() +
  facet_wrap(~hyperactivity) +
  scale_y_continuous(breaks = 1:13) +
  theme_classic() +
  labs(x = "Game type", y = "# of Violent Images",
       fill = "Hyperactivity")
```



And an effect plot of the means

```
games_final |>
  ggplot(aes(x = game_type, y = nviolent_images,
             color = hyperactivity,
             group = hyperactivity)) +
  stat_summary(geom = "point", fun = mean) +
  stat_summary(geom = "line", fun = mean) +
  theme_classic() +
  labs(x = "Game type", y = "# of Violent Images",
       color = "Hyperactivity")
```



These descriptive statistics reveal that violent video games tend to increase the number of violent images children produce regardless of their hyperactivity.

The effect, though, differs based on hyperactivity: For hyperactive children, the effect of violent video games increasing the number of violent images was far stronger within hyperactive children than non-hyperactive children.

Written out more formally (in APA style):

The type of game tends to increase the number of violent images children produced. Hyperactive children playing prosocial games produce an average of four ($SD = 1.22$) violent images, whereas these children playing violent games produce far more ($M = 11.2$, $SD = 1.30$) violent images.

Among non-hyperactive children, those playing prosocial games produce an average of 4.6 ($SD = 1.14$) violent images, while those playing violent video games tend to produce an average of nine ($SD = 1.58$) violent images.

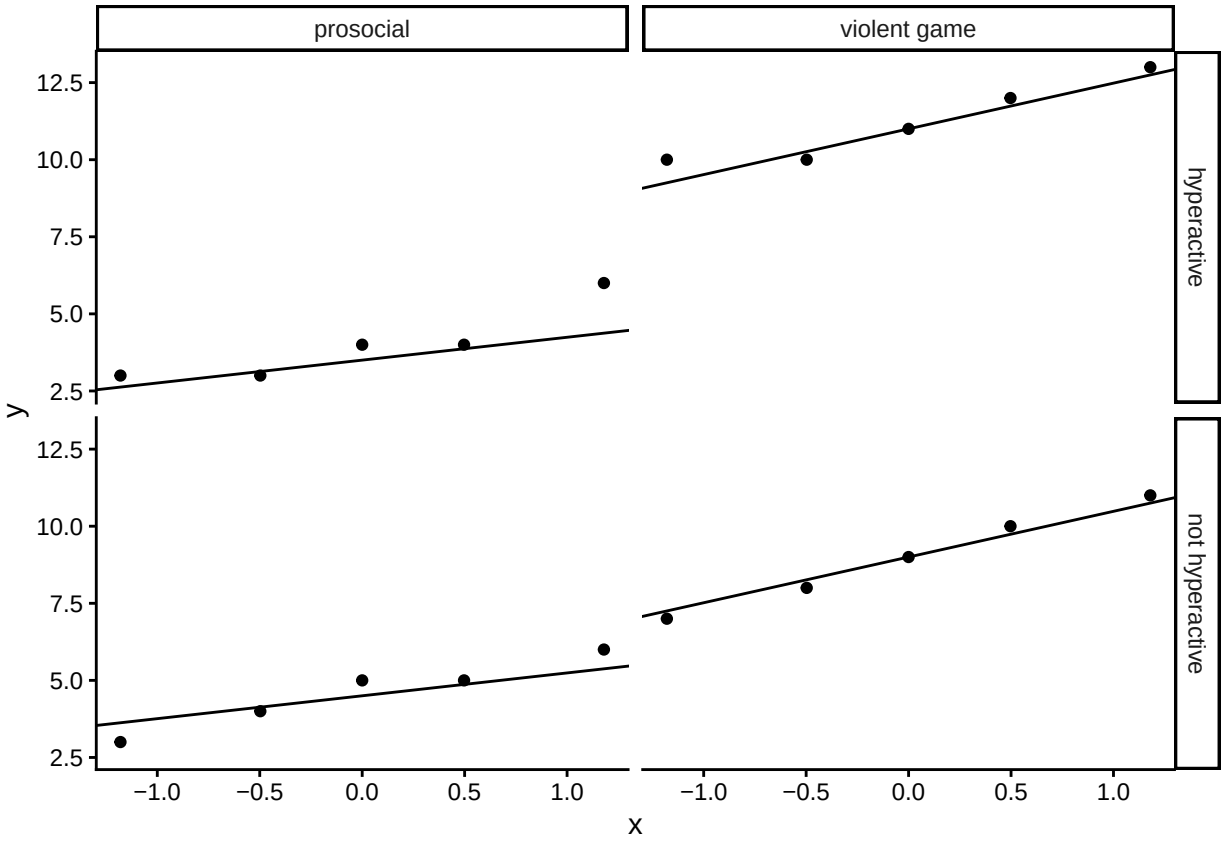
4)

We conduct our two-way ANOVA as follows:

a) Assess assumptions

We start by assessing the normality assumption.

```
games_final |>
  ggplot(aes(sample = nviolent_images)) +
  geom_qq() +
  geom_qq_line() +
  facet_grid(hyperactivity ~ game_type) +
  theme_classic()
```



Based upon the above Q-Q plot, the close alignment of points to the theoretical QQ-lines suggest normality is met.

(You can also assess this with descriptive stats and/or other visualizations, such as density plots or boxplots.)

The other assumption is homogeneity of variances.

```
res <- ezANOVA(games_final,
               wid = ID,
               between = hyperactivity:game_type,
               dv = nviolent_images,
               return_aov = TRUE)
```

```
## Coefficient covariances computed by hccm()
```

```
res$`Levene's Test for Homogeneity of Variance`
```

```
##   DFn DFd  SSn  SSd         F         p p<.05
## 1   3  16 0.55 10.4 0.2820513 0.8375836
```

The assumption of homogeneity of variances is also met! (I wonder if these data were simulated. . .)

We will implicitly assume that the other assumptions (e.g., independence, random sampling from a homogenous population) are both met.

b) Conduct the ANOVA

```
res$ANOVA
```

```
##                     Effect DFn DFd         F            p p<.05       ges
## 1           hyperactivity   1  16  1.828571 1.951015e-01       0.1025641
## 2               game_type   1  16 96.114286 3.616344e-08     * 0.8572885
## 3 hyperactivity:game_type   1  16  5.600000 3.091323e-02     * 0.2592593
```

There is a clear instnace of an interaction effect: the effect of hyperactivity on the number of violent images differes as a function of the nature of the video game children played.

Post-hoc tests:

```
mod_1 <- aov(nviolent_images ~ hyperactivity*game_type,
             data = games_final)
TukeyHSD(mod_1)$`hyperactivity:game_type` |>
  as.data.frame() |>
  # Arrange results from most positive to most negative
  arrange(desc(diff)) |>
  # Indicate which means are statistically significant
  mutate(is_sig = `p adj` < .05)
```

```
##                                                diff       lwr       upr
## hyperactive:violent game-hyperactive:prosocial          7.2  4.806299 9.5937009
## hyperactive:violent game-not hyperactive:prosocial      6.6  4.206299 8.9937009
## not hyperactive:violent game-hyperactive:prosocial      5.0  2.606299 7.3937009
## not hyperactive:violent game-not hyperactive:prosocial  4.4  2.006299 6.7937009
## not hyperactive:prosocial-hyperactive:prosocial         0.6 -1.793701 2.9937009
## not hyperactive:violent game-hyperactive:violent game  -2.2 -4.593701 0.1937009
##                                                             p adj is_sig
## hyperactive:violent game-hyperactive:prosocial         1.171886e-06   TRUE
## hyperactive:violent game-not hyperactive:prosocial     3.647740e-06   TRUE
## not hyperactive:violent game-hyperactive:prosocial     1.036017e-04   TRUE
## not hyperactive:violent game-not hyperactive:prosocial 4.093028e-04   TRUE
## not hyperactive:prosocial-hyperactive:prosocial        8.888372e-01  FALSE
## not hyperactive:violent game-hyperactive:violent game  7.736869e-02  FALSE
```

The post-hoc tests confirm the interaction effect we found earlier.

5)

I ran a two-way ANOVA to determine whether there was a statistically significant interaction effect between hyperactivity (hyperactive vs. not) and whether children played violent or prosocial video games on the number of violent images they generated.

Descriptively, the type of game tended to increase the number of violent images children produced. In particular, hyperactive children playing pro-social games produced an average of four ($SD = 1.22$) violent

images, whereas these children playing violent games produced far more ($M = 11.2$, $SD = 1.30$) violent images.

Among non-hyperactive children, children playing prosocial games produced an average of 4.6 ($SD = 1.14$) violent images, while those playing violent video games produced an average of nine ($SD = 1.58$) violent images.

Prior to running the ANOVA, I checked the assumptions of normality and homogeneity of variance, and confirmed that these assumptions were met with close alignment of observed data to Q-Q line and a statistically non-significant Levene's test ($F(3, 16) = 0.28$, $p = .838$), respectively. The assumptions of the independence of observations and random selection were both assumed.

The two-way ANOVA showed a statistically significant interaction effect, $F(1, 16) = 5.60$, $p = .03$, $ges = 0.259$.

Tukey's honestly significant difference tests were conducted on all of the mean differences. Most mean differences were statistically significant, except for the mean differences between children who had a different hyperactivity level but played the same type of video game.

To illustrate the interaction, hyperactive children playing violent games generated 7.2 more violent images than those playing prosocial games ($p < .001$, 95% CI [4.81, 9.59]). Non-hyperactive children playing violent video games generated 4.4 more violent images than those playing a prosocial game ($p < .001$, 95% CI [2.01, 6.79]).

In conclusion, playing violent video games corresponded with a statistically significant increase in the number of violent images children generated, and this difference was stronger for hyperactive children than non-hyperactive children.